

Chemical etching of Cu-ETP copper

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Abstract

Chemical etching is the controlled dissolution of workpiece material by contact with strong chemical solution. The process can be applied to any material. Copper is one of the extensively used engineering material in the fabrication of microelectronic components, microengineered structures and precision parts by using chemical etching process. In this study, copper is chemically etched with two different etchants (ferric chloride and cupric chloride) at 50 °C. The effects of selected etchants and machining conditions on the depth of etch and surface roughness were investigated. The experimental study provided that ferric chloride produced the fastest chemical etch rate, but cupric chloride produced the smoothest surface quality.

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1. Introduction

Advancement in the manufacturing processes has been concentrated in the use of new tools and energy forms. These developments produced different new machining processes than conventional machining processes and they were named “nontraditional machining processes”. They have been extensively and successfully used since 1950s.

Nontraditional machining processes are widely employed to manufacture geometrically complex and high-dimensional accurate machine parts from advanced materials in industries as diverse as aerospace, electronics and automotive. Moreover, various nontraditional machining processes have been used in micron-size parts production. The trend seems increasing in future.

Chemical etching is one of the oldest nontraditional machining processes. It uses strong chemical solution, which is called etchant, to remove unwanted workpiece material by controlled dissolution. The process is widely used in electronics, precision engineering and medical industries to produce micro-components. It is also a useful process in removing material from sheet components to reduce weight. It can be

applied to any engineering material from steel to silicon. The process is sometimes called chemical machining, chemical milling, wet etching, etching, etc. [1].

Chemical etching is not a new process; it is used to shape metals in the ancient Egypt where copper was etched with citric acid to produce jewelry around 2500 B.C. The process replaced hand-tool engraving process in the 15th century, mineral acids were used for weapons, helmets and breastplates. In the 17th century, it was used for the first time as a manufacturing process of steel parts. Developments in the chemistry provided significant advancement, various acid types produced and resists were discovered during the 18th and 19th centuries [2,3].

First patent (British Patent: 565) was taken by William Fox using ferric chloride (FeCl_3) for etching copper. Later, John Baynes described chemical etching of materials from two sides in the US Patent 378,423 in 1988 [3]. The development of the process continued in later years.

Major developments were noticed after the Second World War and the process has been widely applied as manufacturing process since 1950s. First industrial application was conducted by the North American Aviation, Inc. (California, USA) to etch aluminium components for rockets. The company named the process “chemical milling” and patented it (US Patent No: 2,739,047) in 1956. The process was also

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used for removing excess mass from aluminium wing skins and other airframe parts. The advancement in the process resulted wider industrial applications from microelectronics components to medical parts and decorative items. The process has advantages over traditional machining processes are [1–5]:

- a. Required no special tools;
- b. Short necessary machining time;
- c. Fairly economical machining process;
- d. No plastic strain;
- e. Machine complex geometrical shapes with high precision.

Copper, as one of the major commercial engineering materials, is extensively used in various industries such as electronics, automotive and chemical industries because of its excellent electrical and thermal conductivities, good strength and fatigue resistance, high corrosion resistance and ease of fabrication [6].

Copper chemically machined with different etchants for various applications. Mainly FeCl_3 , cupric chloride (CuCl_2) and ammoniacal alkaline etchants have been used. The selection of optimum etchant solution depends on some etchant properties as follow [7,8]:

- a. High machining rate;
- b. Easy control and stable machining process;
- c. Generation of no toxic fumes;
- d. Regeneration of waste etchant and copper recovery;
- e. Environmentally friendly.

FeCl_3 is an universal etchant and can be used for general engineering materials such as steels, aluminium and its alloys, copper and its alloys, nickel, etc. It is very useful etchant for small-scale copper etching process. It can dissolve copper up to approximately 120 g/l if the prolonged chemical etching time is acceptable. Major setback of this etchant is the impossibility to regenerate economically. This practically rules out its use for mass copper etching applications. Copper recovery from the used etchant is difficult and economically not possible. Various copper etching studies conducted with FeCl_3 [4,9–21]. The etching process was high and produced better etching properties.

CuCl_2 provides rapid chemical etching. It offers the advantages of easy regeneration. The copper dissolve capacity is also higher than FeCl_3 , i.e. around 150 g/l in practical applications. It produces no sludge formation as FeCl_3 does. Therefore this etchant seems more economical etchant for copper etching in comparison to FeCl_3 . The literature survey mentioned that although the etch rate of CuCl_2 is lower than FeCl_3 , it is more suitable etchant for copper and copper alloys etching [9–13,15–24].

Ammoniacal alkaline etchants have industrially introduced in the beginning of 1970s. The chemical reaction of these etchants is more complex than FeCl_3 and CuCl_2 etchants. They can dissolve up to 225 g/l copper and provides higher etching rates. Most of the etchants are proprietary formulations and details are hardly available. The stability of

etchant is critical. The etching process is closed-loop system and needs high investment. These etchants can be regenerated and etched copper can be recovered. These etchants are only economical if copper etching process is as mass production [21].

The literature review produced information that FeCl_3 and CuCl_2 are suitable etchants for low and medium production levels. These are also capable to machine copper alloys. Therefore the present study concentrated on these two etchants for chemical etching of copper.

For FeCl_3 etchant applied for copper, the concentration of the etchant solution should be higher than 3 M, otherwise, it is reported, surface finish would be poor. However high etchant concentration decreases etch rate [4].

The use of CuCl_2 etchant presents different properties. The etch rate increases with increasing etchant concentration. Similar trend was also mentioned for surface roughness, high etchant concentration produced a better surface quality [25].

The selection of etching temperature influences etch rate and surface finish. In industrial application of chemical etching of copper, the highest possible etching temperature the etchant machine allowed is used. This value is around 50 °C. Low etching temperature decreases etch rate and produces poor surface quality [25].

In this study, the selected FeCl_3 and CuCl_2 chemical etchants were used to machine copper chemically. The concentrations of these etchants were 3.76 M for FeCl_3 and 2.33 M for CuCl_2 . The selected chemical etching temperature was 50 °C, this value was generally applied for industrial chemical etching process. The experimental study was conducted in beaker. The study concentrated on the parameters of depth of etches, surface roughness and etched copper weights. The influence of etchant solution on the parameters were investigated and compared.

2. Experimental

The selected copper material was electrolytic tough pitch high conductivity copper (ISO code is Cu-ETP) and its chemical composition was 99.90% Cu, 0.005% Pb and 0.001% Bi. The hardness of materials was 55 HV. The thickness of specimens was 1 mm and cut at 20 mm × 100 mm dimension. The specimens were cleaned to remove chemical and particulate contaminations such as oil, wax, grease, rust, oxide and dirt in ultrasonic cleaning machine. The prepared cleaning solution was mixture of distilled water and addition of 1% HCl. The cleaning process was completed at 40 °C for 25 min. The cleaned specimens were rinsed with tap water and subsequently examined by dripping water on the cleaned surfaces. A uniform, continuous thin film of water spreading over the cleaned surfaces indicated that the surface cleaning was adequate.

FeCl_3 and CuCl_2 chemical etchants were prepared at 3.76 and 2.30 M, respectively. The amount of etchant for each experiment was 100 ml. Etchant was poured into beaker that

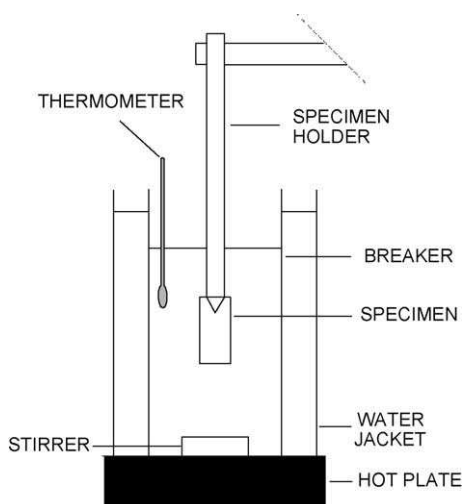


Fig. 1. Experimental set-up of chemical etching in beaker.

was put into water jacket to control chemical etching temperature (Fig. 1). The selected machining temperature was $50 \pm 2^\circ\text{C}$. Double-sided chemical etching process was conducted. The chemical etching period was 25 min in total and the measurements were completed after 5-min intervals.

The measurements of thickness were carried out by Mitutoyo outside micrometer (deviation was ± 0.001) and surface roughness was measured by Taylor-Hobson Surtronic 3+. For each etchant, three specimens were chemically machined and three different measurements were taken from each specimen.

3. Experimental results and discussion

It is important to find the chemical reaction of selected copper material with etchant. Each etchant reacts differently during chemical etching process. The chemical reaction of copper with FeCl_3 is as follows:



Copper is oxidized by the ferric ions, forming cuprous chloride (CuCl) and ferrous chloride (FeCl_2). CuCl oxidized further in the etchant solution to produce CuCl_2 as follows:



The built-up CuCl_2 itself reacts also with copper and forms CuCl shown below:



The chemical reaction of CuCl_2 with copper is simpler. The copper surface gets attacked by CuCl_2 while CuCl is formed. The etchant solution is not effective in further chemical etching process.

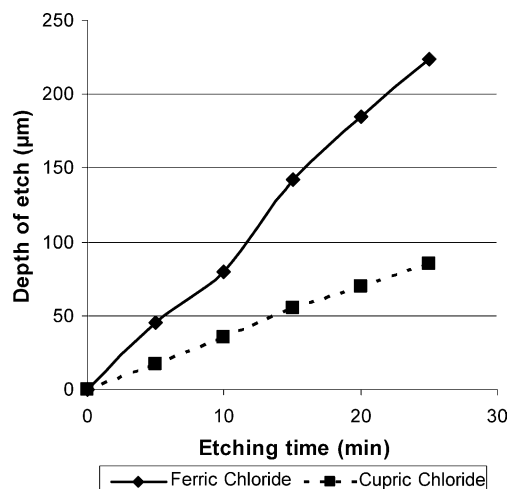


Fig. 2. Depth of etch against etching time.

It can be noticed that using FeCl_3 etchant for chemical etching of copper produces complex chemical reaction and provides Fe^{2+} , Cu^{2+} and Cu^+ , but using CuCl_2 only produces Cu^+ . This is a major factor for the regeneration of spent etchant after chemical etching process. It is a general approach that regeneration of any spent etchant is easy and economical if the spent etchant is simple. From this point, CuCl_2 seems more suitable etchant for chemical etching of copper.

The depth of etch was examined and the results were shown in Fig. 2. It was observed that the use of FeCl_3 produced the highest value in comparison to CuCl_2 etchant. After 25-min etching period, depths of etch value were $224 \mu\text{m}$ for FeCl_3 and $85 \mu\text{m}$ for CuCl_2 etchants. The chemical etch rate is $9.3 \mu\text{m}/\text{min}$ for FeCl_3 and $3.4 \mu\text{m}/\text{min}$ for CuCl_2 .

FeCl_3 etchant in copper machining behaves like two etchants in the process. The main etchant is FeCl_3 and the byproduct etchant is CuCl_2 . This causes a higher etch rate in comparison to CuCl_2 etchant only.

The examination of surface roughness for each etchant was conducted and the results were given in Fig. 3. General expectation from chemical etching process is to obtain the best possible surface finish. This is greatly important in the weight reduction of material. It was observed that FeCl_3 produced a rougher surface roughness than CuCl_2 .

The surface roughness decreased in the beginning of chemical etching process and increased during etching process in FeCl_3 application. After 15-min etching period, the surface roughness value became stable. The surface roughness of R_a was $1.92 \mu\text{m}$. The surface roughness presented different trend for CuCl_2 ; surface quality increased for first 10-min etching and then the surface roughness slightly increased and became constant. The value of R_a was $0.73 \mu\text{m}$ after 25 min of etching period.

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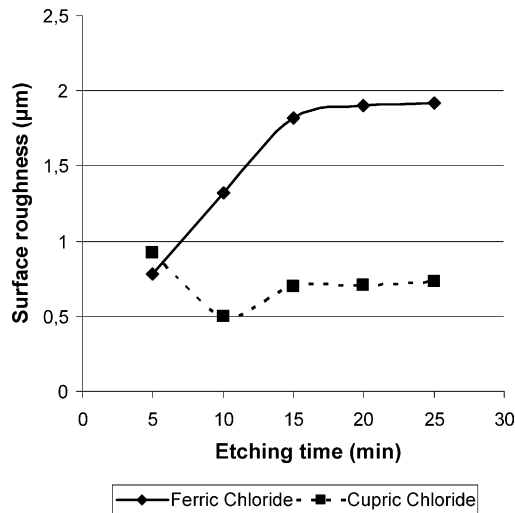


Fig. 3. Effect of etchants on surface roughness.

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The other selected chemical etching parameter was the amount of etched copper weight. It was noticed that FeCl_3 dissolved the highest amount of copper after etching period. This weight was 1.5 g (15 g/l). The value of etched copper weight was halved when CuCl_2 was used for copper etching. The result of etched copper weights for both etchants was shown in Fig. 4.

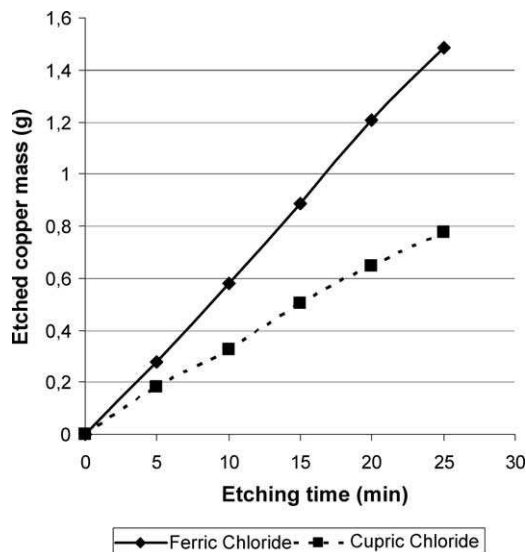


Fig. 4. The amount of etched copper mass vs. etching time.

4. Conclusion

The present study investigated two widely used etchants (FeCl_3 and CuCl_2) for chemical etching of copper. The experimental results can be concluded as follow:

- The chemical etch rate is important factor for any material machining. The highest chemical etch rate obtained with using FeCl_3 . This would make the machining time of process higher.
- Surface roughness is another factor to be considered especially in case of weight reduction of material. CuCl_2 produced the smoothest surface quality comparing to FeCl_3 .
- The dissolved copper amount was compared and it was obtained that FeCl_3 etched more copper than CuCl_2 for the same etchant volume and etching period. This means that the prolonged etching time would be longer for FeCl_3 etchant used for copper etching.
- Chemical reactions of copper with selected etchants were examined and CuCl_2 is resulted a simpler reaction than FeCl_3 . This conclusion is effective parameter for regeneration process of used etchant. This would make regeneration process easier and more economical.

The experimental study of chemical etching of copper examined basic etching parameters. Next step of this study will be investigating other parameters such as the pH level and oxidation redox potential of used etchants, the influences of chemical additives to the main etchant.

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